

Guided Capstone Project Report

By Parvati Neelakantan

1. Executive Summary

Big Mountain Resort recently invested in a new chair lift, increasing annual operating costs by \$1.54 million. Management wants to know whether the resort's current ticket price reflects the value of its facilities and identify opportunities to increase revenue while remaining competitive.

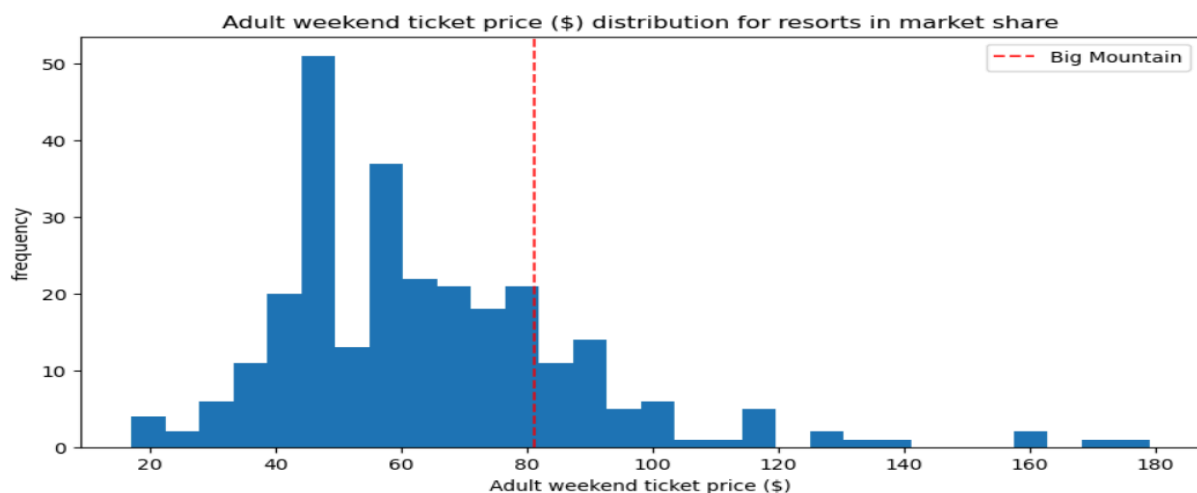
Using data from 276 comparable ski resorts, I developed and evaluated several machine learning models to predict ticket prices based on resort features. The Random Forest model performed best and estimated that Big Mountain's facilities support an adult weekend ticket price of **\$95.87**, compared with the current price of **\$81.00**. Even after accounting for the model's average prediction error ($\pm \$10.39$), the results suggest the resort is likely underpriced.

Based on these findings, I recommend a gradual ticket price increase rather than an immediate adjustment to the predicted price. The analysis also shows that the recent chair lift investment is financially justified and provides a practical framework for evaluating future pricing and investment decisions.

2. Business Problem

Big Mountain Resort recently invested in a new chair lift, increasing annual operating costs by \$1.54 million. Management wants to determine whether the current adult weekend ticket price reflects the value of the resort's facilities and identify opportunities to increase revenue while remaining competitive. The goal is to support pricing and investment decisions using data rather than intuition.

Figure 1: Big Mountain Resort is priced below the peer resort distribution, indicating potential upward pricing headroom



3. Data Preparation

The dataset contained information from 276 ski resorts after cleaning and preprocessing. Missing values were handled where necessary, categorical variables were encoded, and additional features such as state-level metrics were engineered. Big Mountain Resort was excluded during model training and used only for final prediction to avoid bias.

4. Exploratory Data Analysis

Exploratory analysis showed that ticket price generally increases with resort size and infrastructure. Features such as vertical drop, number of runs, snowmaking acreage, and fast quads showed the strongest relationships with ticket price and were therefore expected to be important predictors.

5. Key Findings

Model	CV MAE (\$)	Test MAE (\$)	R ²
Dummy Regressor (mean baseline)	–	19.00	0.00 (slightly negative on test)
Linear Regression (all features)	–	9.41	0.82 train / 0.72 test
Linear Regression (best, k=8)	10.50 ± 1.62	11.79	0.70 (CV)
Random Forest (best, 69 trees)	9.64 ± 1.35	9.54	0.71 (CV)

5.1 Baseline Benchmark

As a benchmark, I used a Dummy Regressor that always predicts the average ticket price. It achieved a test MAE of about **\$19**, providing the minimum performance that any predictive model should exceed.

5.2 Linear Regression Results

The Linear Regression model explained **72%** of the variation in ticket prices on the test set and achieved a test MAE of **\$11.79**. Feature selection improved the model by reducing overfitting and retaining the eight most informative features.

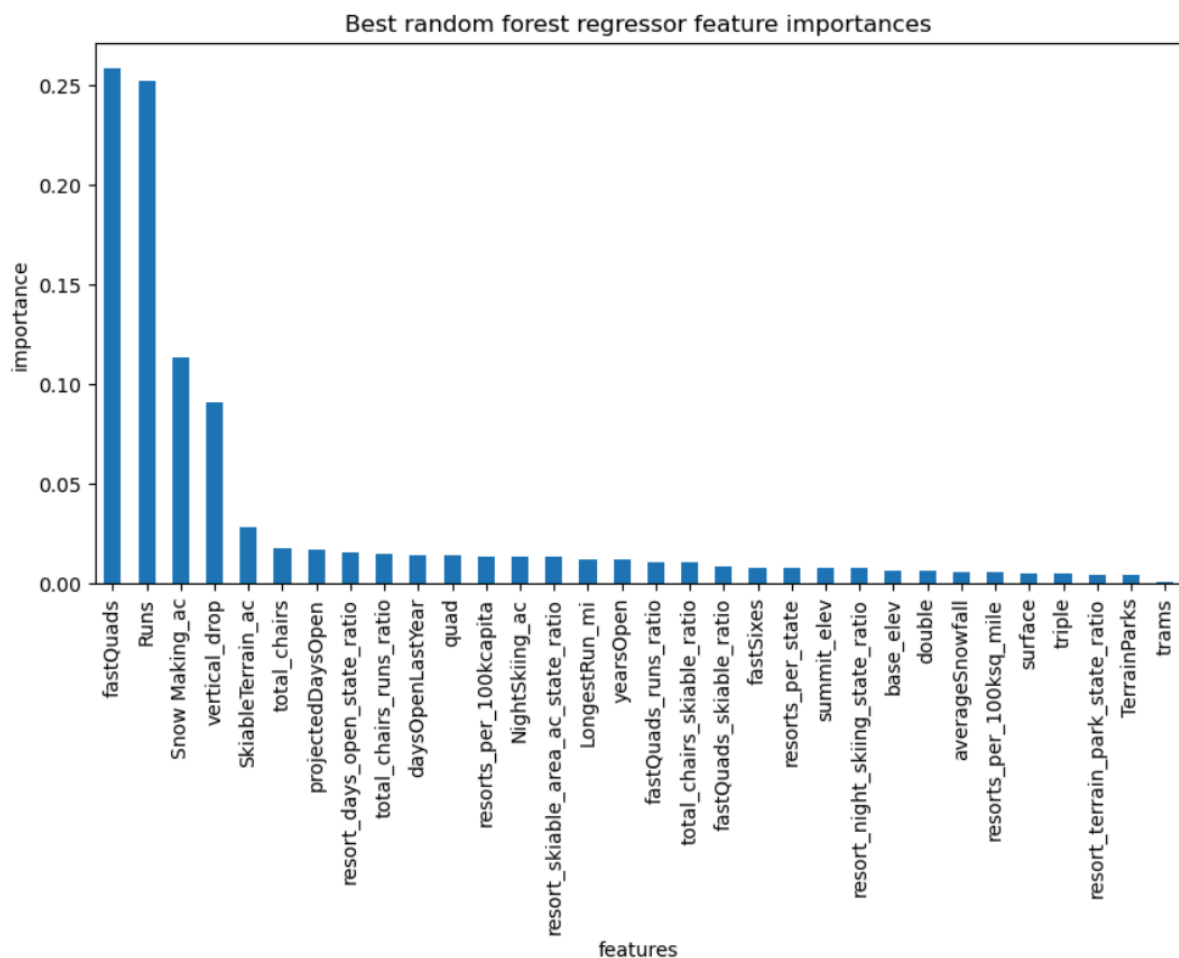
The strongest positive drivers of ticket price were **vertical drop, snowmaking acreage, total chairs, fast quads, number of runs, and longest run**. Cross-validation produced an MAE of **\$10.50 ± \$1.62**, which was close to the test error, indicating reliable performance.

5.3 Random Forest Results

The Random Forest model produced the best overall performance, with a cross-validated MAE of **\$9.64 ± \$1.35** and a test MAE of **\$9.54**. The close agreement between cross-validation and test results suggests the model generalizes well to new data.

The most influential features were **fast quads**, **number of runs**, **snowmaking acreage**, and **vertical drop**, closely matching the findings from the Linear Regression model.

Figure 2. Random Forest Feature Importance



5.4 Model Performance Comparison

The learning curve shows that model performance improves rapidly with additional training data before leveling off after about **40–50** training examples. This suggests the current dataset is sufficient, and collecting substantially more data is unlikely to produce large improvements in predictive accuracy.

Figure 3: Learning curve



5.5 Final Model Selection

The Random Forest model was selected as the final model because it produced the lowest prediction error, showed consistent performance across cross-validation and the test set, and identified the same key pricing factors as the Linear Regression model. These consistent findings increase confidence that the identified resort features are the main drivers of ticket price.

6. Recommendations

I conducted four scenario analyses. The first examined the effect of closing up to 10 of the least-used runs on ticket price and revenue. The second evaluated the impact of adding one run, increasing the vertical drop by 150 feet, and installing an additional chair lift. The third repeated the second scenario while also adding 2 acres of snowmaking to assess its effect on ticket price and revenue. The final scenario increased the longest run by 0.2 miles and guaranteed snow coverage by adding 4 acres of snowmaking capacity. Based on the results of these analyses, I present the following recommendations.

6.1 Gradually increase the ticket price

The model predicts that Big Mountain's facilities support an adult weekend ticket price of **\$95.87**, compared with the current price of **\$81.00**. Rather than increasing the price immediately, I recommend a gradual increase while monitoring ticket sales and visitor numbers. This approach allows the resort to capture additional revenue while reducing the risk of lowering demand.

6.2 Continue investing in high-value infrastructure

The new chair lift appears to be a worthwhile investment. The additional annual operating cost of **\$1.54 million** is equivalent to about **\$0.88 per ticket**, assuming annual sales of **1.75**

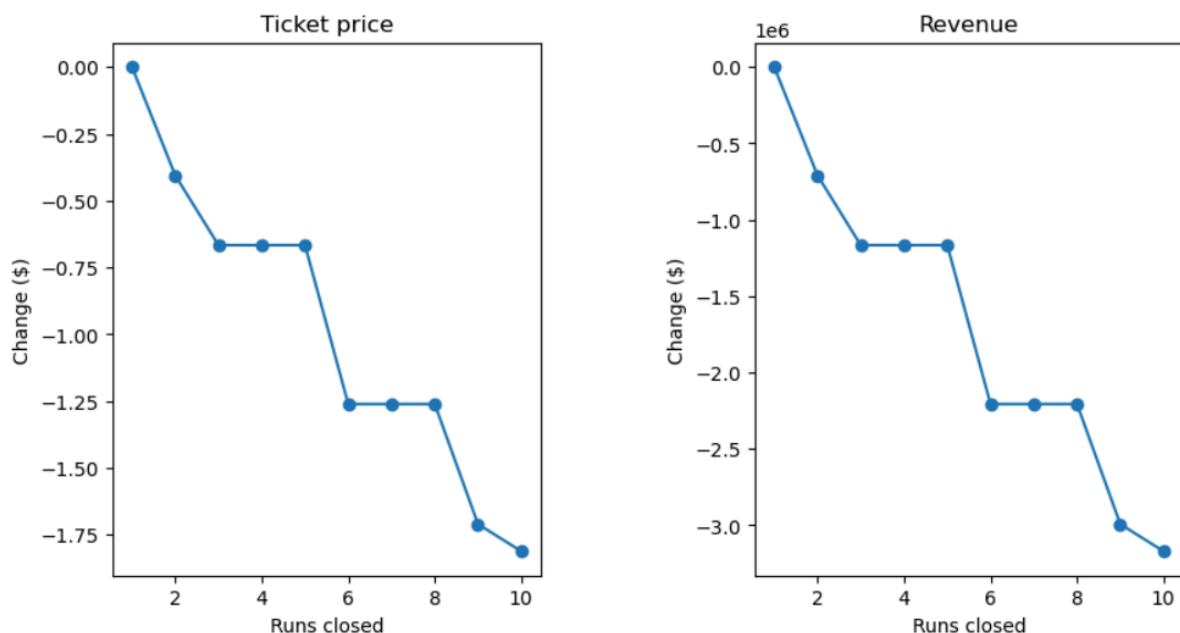
million tickets. The model estimates that the improvements included in **Scenario 2** support a **\$1.99** increase in ticket price, which would generate approximately **\$3.47 million** in additional annual revenue. This suggests the investment can more than cover its operating cost.

6.3 Prioritize Scenario 2 over the other investment options

Of the four scenarios evaluated, **Scenario 2** provides the greatest financial benefit. Adding one run, increasing the vertical drop by **150 feet**, and installing the new chair lift increases the supported ticket price without unnecessary additional investment. Expanding snowmaking capacity (Scenario 3) and the changes in Scenario 4 do not increase the supported ticket price further, so they are unlikely to provide sufficient returns based on the current model.

6.4 Evaluate run closures carefully

Closing one run has little effect on the supported ticket price, and closing up to five runs results in only a small reduction. However, closing six or more runs leads to a noticeable decrease in the predicted ticket price. If reducing operating costs becomes necessary, I recommend temporarily closing the least-used run and monitoring visitor numbers and ticket sales before making permanent changes.



7. Limitations & Future Work

The main limitation of this analysis is the lack of cost data. Since ticket price is the only financial variable available, the model can estimate potential revenue but not profitability. Additional data on operating, maintenance, labor, and capital costs would allow future analyses to evaluate the financial impact of investment decisions more accurately.

The model predicts that Big Mountain could support a higher ticket price because it compares the resort with others that offer similar facilities. However, factors such as brand reputation, local competition, customer preferences, and management's pricing strategy are not included in the data and may influence the optimal ticket price.

If adopted by the resort, the model could be developed into a simple decision-support tool that allows managers to evaluate pricing and investment scenarios. Updating the model with new data each season would help maintain its accuracy over time.

8. Expected Business Impact

Implementing these recommendations could increase annual revenue while maintaining Big Mountain's competitive position. A gradual ticket price increase would better align prices with the value offered by the resort while reducing the risk of negatively affecting demand.

The analysis also provides a data-driven framework for evaluating future investments. Rather than relying solely on experience or intuition, management can use the model to estimate how changes to resort facilities may influence ticket prices and revenue before committing to major capital projects.

9. Conclusion

This project used machine learning to evaluate whether Big Mountain Resort's current ticket price reflects the value of its facilities. The results suggest the resort is likely underpriced and that a gradual price increase, together with the recent chair lift investment, could increase revenue while maintaining competitiveness.

More broadly, the project demonstrates how predictive analytics can support pricing and investment decisions, helping management make more informed, data-driven choices.